# The MEW nozzle in two dimensions

First results for the set-up of Ju, Omar, Mirihanage and Kilic

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**Summary.** The set-up of *Suppressing oozing during filament-based melt electrowriting* (SSRN 7582986) was simulated as a planar slice through the nozzle: PCL at 120 °C fed through a 0.4 mm nozzle at 6.5 kV, 5.5 mm above a collector moving at 2100 mm min$^{-1}$. Four results:

- At the feed printed in Table 1, 0.255 mm$^3$ s$^{-1}$, the melt is pushed out as a column whether the field is on or off.
- At 0.255 mm$^3$ min$^{-1}$ the field shapes a meniscus that grows at exactly the feed rate; no jet can form on this grid.
- Only the per-minute feed matches the measured fibres: at 35 mm s$^{-1}$ it lays a 12 µm fibre.
- After the melt reaches the collector the model is not reliable, and its absolute numbers are not grid-converged. Comparisons at one resolution, such as field on against field off, are what it supports.

## 1 Model and set-up

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| Melt | PCL held at 120 °C. Viscosity 500 Pa s and surface tension 0.03 N m$^{-1}$ are nominal values, not measurements; density 1080 kg m$^{-3}$. |
| Nozzle | 0.4 mm bore, as in the paper; an assumed 1.0 mm tip face, 5.5 mm above the collector. |
| Field | 6.5 kV, and 12 kV in one run. The melt conducts within 2 mm of the tip and is an insulator below that, so it cannot short the gap. |
| Collector | 35 mm s$^{-1}$ (2100 mm min$^{-1}$). |
| Feed | 0.255 mm$^3$ s$^{-1}$ as printed in Table 1, so the melt leaves the bore at 2.0 mm s$^{-1}$; and 0.255 mm$^3$ min$^{-1}$. |
| Grid | 25 µm cells over an $8 \times 6.8$ mm slice, 14 µs steps. A free-surface lattice Boltzmann model of the melt (D2Q9, central moments; the air is not simulated), and Laplace's equation for the potential in the air (D2Q5). The melt surface feels the electric pull $\varepsilon_0 E^2/2$. |

Matched to the experiment: the capillary numbers of the feed (34 as printed, 0.56 per minute) and of the collector (583), the Bond number (0.014), and the ratio of electric to capillary pressure, $\varepsilon_0 V^2/\gamma R = 62$ at 6.5 kV. Not matched: inertia, which is negligible in the experiment and kept small in the model (Reynolds number at most 0.32).

## 2 Results

### 2.1 The printed feed: a column, field or no field

The capillary number in the bore is 34, so viscous extrusion beats surface tension by a wide margin. With the field on and off the front falls at 1.13 mm s$^{-1}$, the melt within 1 mm of the tip settles at 0.65 mm$^2$ in both, and both columns reach the collector at 4.7 s, within 0.01 s of each other (Fig. 1). In the experiment this would look like oozing whatever the voltage.

### 2.2 The per-minute feed: a meniscus that grows at the feed rate

At 6.5 kV and at 12 kV alike the melt under the tip grows at 0.013 54 mm$^2$ s$^{-1}$, against a feed of 0.013 53 mm$^2$ s$^{-1}$ in this planar model: nothing carries melt away (Fig. 2). A planar sheet carrying this feed at 35 mm s$^{-1}$ would be 0.4 µm thick, against 25 µm cells, so no jet can form on this grid. After 8 s the meniscus reaches 0.32 mm below the tip face at 6.5 kV and 0.35 mm at 12 kV, and the strongest pull on it is 0.68 and 2.51 times the bore's capillary pressure, $\gamma/R = 150$ Pa. A flat tip concentrates the field less than a round one; the 12 kV run estimates the pull a round 1 mm tip feels at 6.5 kV.

### 2.3 Which feed?

The paper's fibres are 9.5 µm to 13.5 µm with the fan cooling and about 19 µm to 22 µm after nozzle cleaning. A straight fibre laid at 35 mm s$^{-1}$ carries its cross-section times that speed, so 9.5 µm to 22 µm fibres carry 0.0025 mm$^3$ s$^{-1}$ to 0.013 mm$^3$ s$^{-1}$. 0.255 mm$^3$ min$^{-1}$ is 0.0043 mm$^3$ s$^{-1}$, a 12 µm fibre; 0.255 mm$^3$ s$^{-1}$ would need a 96 µm fibre.

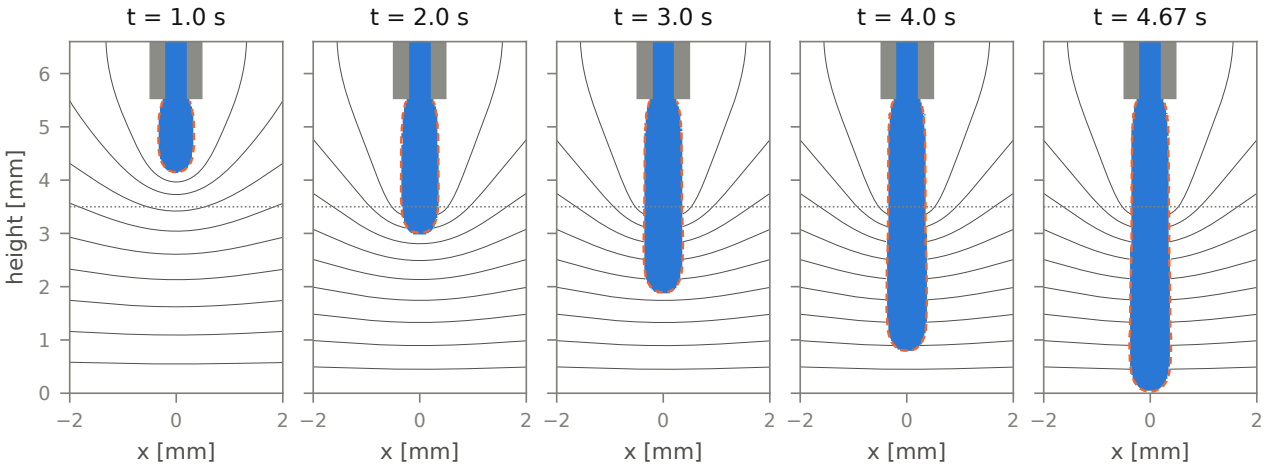



**Figure 1.** Table 1's feed, 0.255 $mm^3\ s^{-1}$, from the start to first contact. Blue: the melt with the field on, equipotentials every 10 % of the voltage, the melt conducting above the dotted line. Orange, dashed: the outline of the melt with the field off at the same instant. The two are almost identical.

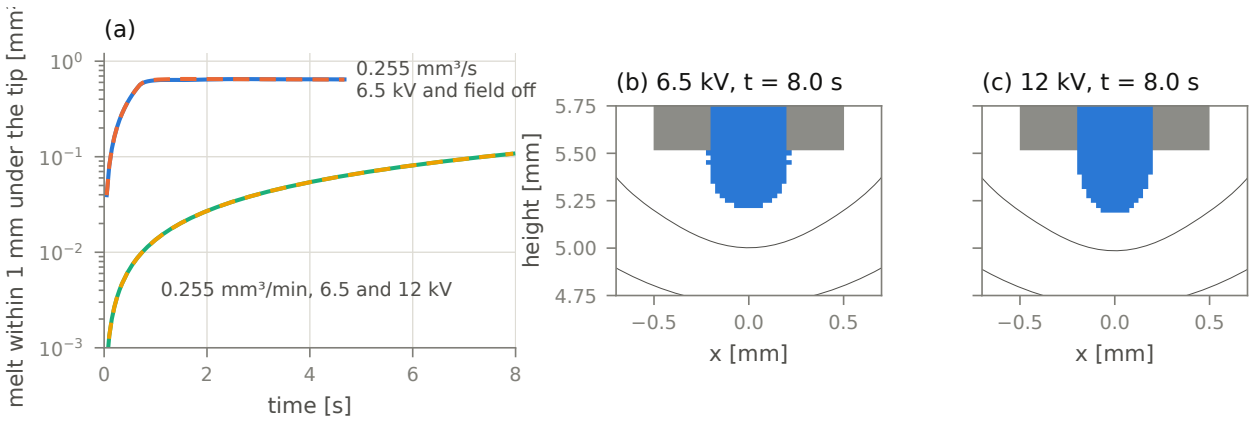



**Figure 2.** (a) Melt within 1 mm under the tip, the model's measure of the accumulation the paper reports as oozing. Printed feed with the field on (blue) and off (orange, dashed), to first contact; per-minute feed at 6.5 kV (aqua) and 12 kV (yellow, dashed). The pairs overlap. (b, c) The per-minute meniscus after 8 s, with equipotentials every 10 % of the voltage.

### 2.4 After the melt reaches the collector

Here the model stops being a measurement. Both ropes leave the collector within 0.1 s of touching it. With the field off the rope sags back, touches again at 5.87 s near the edge of the box and is laid there, until the drawn thread is thinner than a cell at 7.3 s. With the field on it curls into a hook and does not come back during the run (Fig. 3). The field is strongest where the model stops treating the melt as a conductor, which made that line the first suspect; but with the melt conducting to 1, 2 or 3 mm below the tip the rope rises 2.19, 2.31 and 2.06 mm, so the line is not the cause. Why the field keeps the rope up is not yet known. Where the rope touches down is set by the width of the box, not by the physics.

## 3 What the model leaves out

- **The Taylor cone and the jet.** A planar meniscus has no conical equilibrium, and the jet is far below the grid. A flat tip also concentrates the field less than a round one.
- **Wetting.** The nozzle and the collector have no contact-angle law.
- **Charge and conduction.** The melt is a perfect conductor near the tip and an insulator below; there is no free charge and no current.
- **Heat.** The melt is held at 120 °C; the filament's melting in the hot end, where the fan acts, was modelled separately.
- **Rheology.** Newtonian at a nominal 500 Pa s: no shear-thinning, no elasticity, no die swell.

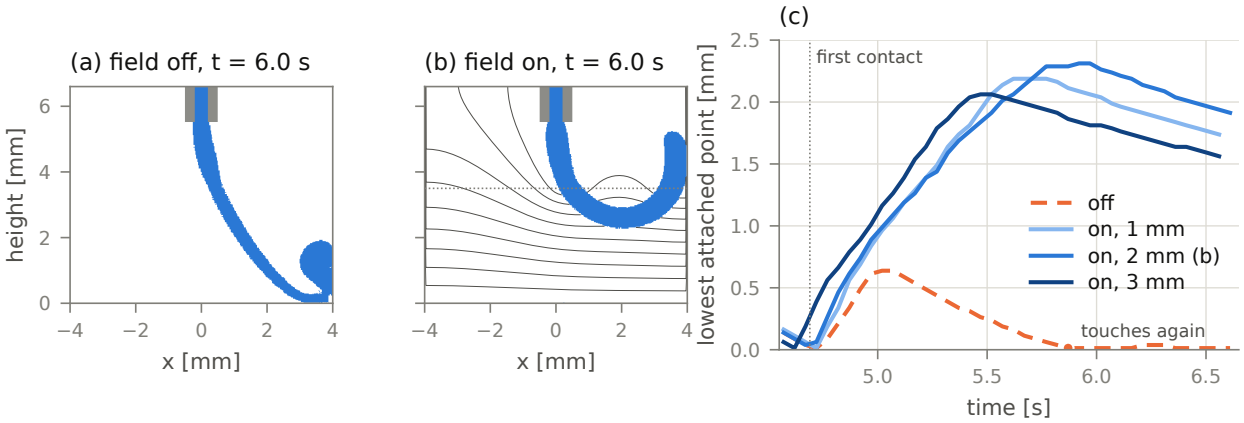



**Figure 3.** After first contact at the printed feed; the collector moves to the right. (a, b) The rope at 6.0 s: field off, it has come back and is laid at the edge of the box; field on, it has curled into a hook. (c) Height of the lowest point of the melt still attached to the nozzle: field off (orange, dashed), and field on with the melt conducting to 1, 2 and 3 mm below the tip (light to dark blue).

## 4 How far to trust it

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| Electric pull on a melt surface, against exact solutions | The pattern of the pull is right to 0.05 %; with the surface free to move, the response is right to 3 % once the surface's own stiffness (next row) is allowed for. |
| A gentle pull on a flat melt surface | Deflections of one or two cells come out 9 % to 25 % short: the free surface is too stiff there, wherever the field moves it by only a cell or two. |
| Surface tension: Laplace's law for a still drop | Right to 1.2 % for drops of 12 cells radius or more, 3.4 % at 8 cells. The bore is 8 cells from wall to axis. |
| Halving the cell size, 50 to 25 µm | The front's speed went from 0.89 to 1.13 mm $s^{-1}$, the melt under the tip from 0.81 to 0.65 $mm^2$, and first contact from 5.9 to 4.7 s. Not converged: compare runs at one resolution rather than quoting these absolute values. After contact the coarse grid shows no effect of the field at all. |
| Mass | Conserved to 0.6 % until contact; it gains up to 5.4 % while the field-off thread is drawn thin, which is not yet explained. |
| Symmetry | The set-up is mirror-symmetric, and the melt is held so until it touches the collector. Without that, round-off pushes the column up to 0.4 mm sideways. |

## 5 Questions for the experiment

1. Is Table 1's feed 0.255 $mm^3$ $s^{-1}$ or 0.255 $mm^3$ $min^{-1}$? At 35 mm $s^{-1}$ the per-minute value lays a 12 µm fibre, inside the 9.5 µm to 13.5 µm measured with the fan; the per-second value would need a 96 µm fibre.
2. Are there measured values for this PCL at 120 °C: the viscosity, ideally a flow curve, and the surface tension? The model uses 500 Pa s and 0.03 N $m^{-1}$.
3. Does the melt ooze with the voltage off? At the printed feed the model pushes out the same column with or without the field.

Simulated with M3LB, a lattice Boltzmann code (github.com/alexderosis/M3LB, branch `electrowriting`): `demonstrator/mew_nozzle` with `-feed s` or `-feed min`, and `-V 12`, `-nofield` or `-lc 1`, `-lc 3` for the variants. Double precision. Figures: `paper/mew_nozzle/make_figs.py`.